



موهبة الإثرائي
البحثي 2026
جامعة الملك فهد للبترول والمعادن



موهبة
Mawhiba
مؤسسة الملك عبدالعزيز
ورجاله للموهبة والإبداع
KING ABDULAZIZ AND HIS COMPANIONS
FOUNDATION FOR GIFTEDNESS AND CREATIVITY



MREP 2026

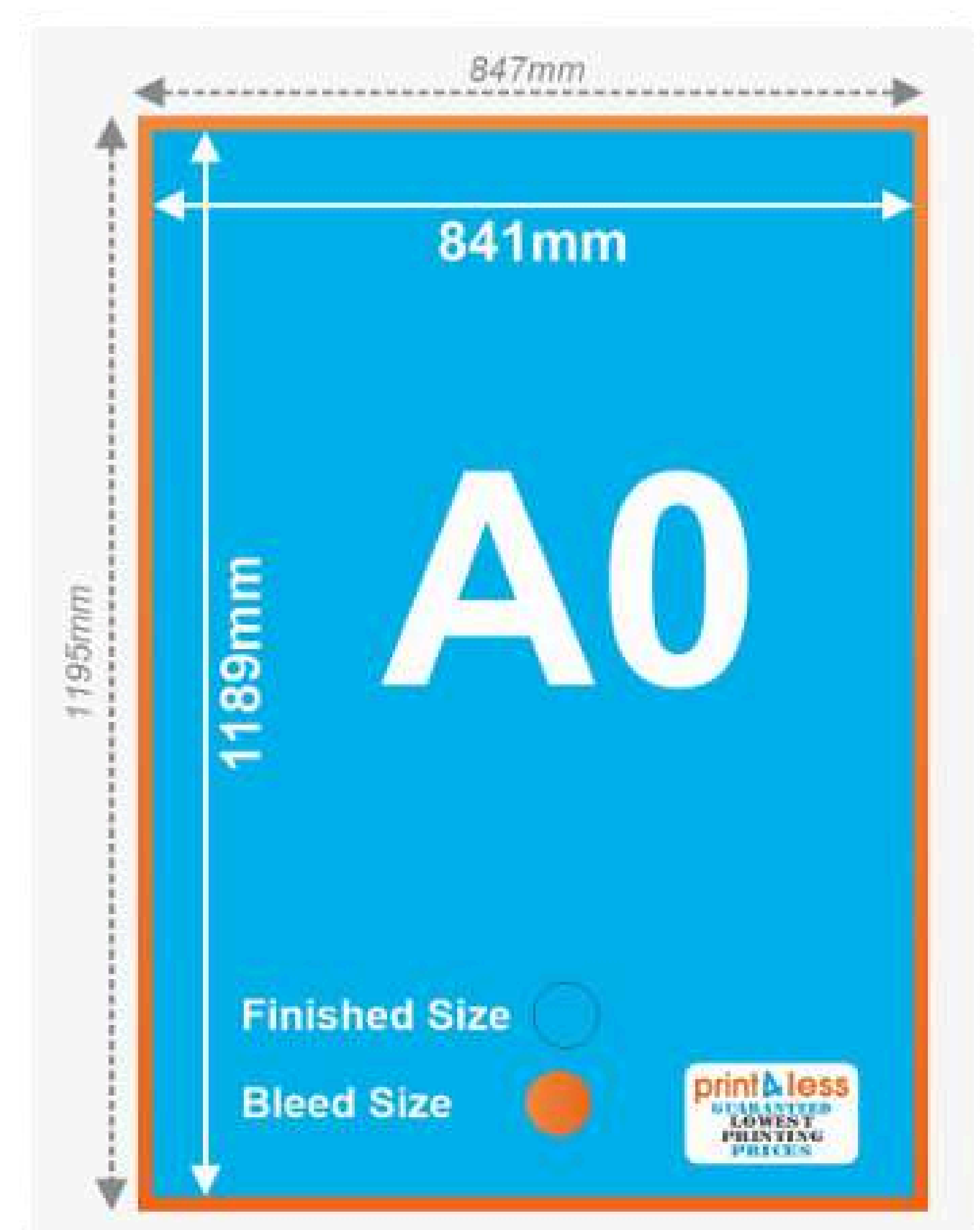
POSTER Guidelines

Measurements & Instructions

A0 Poster

Font Size

- **Title:** 85-100 pt
- **Headings:** 56 pt
- **Main Body Text:** 34 pt
- **Captions and References:** 24 pt



The following must be included on your poster:

- Mohiba logo (on the left)
- KFUPM logo (on the right)
- Program's Logo
- Student's name
- Name of the advisor
- Name of the research center/department.

Logos Needed

موهبة
Mawhiba

مؤسسة الملك عبدالعزيز
ورجاله للموهبة والإبداع
KING ABDULAZIZ AND HIS COMPANIONS
FOUNDATION FOR GIFTEDNESS AND CREATIVITY



موهبة الإثرائي
البحثي 2026

جامعة الملك فهد للبترول والمعادن



صيف
موهبة

- The program's logo may differ
- further changes will be added to the template

Important Sections' Instructions

1. Title



MREP2026_Scientific Poster Template_Energy and Industry

2. Motivation/Introduction

- Should be the purpose / the main story of your project
- Preferable to be 4-5 bullet points
- should contain main statistics that refer to your project problem
- should end into a research gap

Example:

- Global batteries energy storage demand expected to reach 4,700 GWh by 2030. [1]
- Lithium-ion batteries rely on scarce and expensive materials.
- Sodium Ion Batteries (SIBs) utilize sodium, which makes about 2.74% of the earth's crust, lowering its cost significantly. [2]
- While optimal cathodes exist, the challenge lies in **identifying anodes** with sufficient Na^+ intercalation space.
- **Biomass-derived hard carbon (HC)** shows great promise in many fields, especially with a **90%** reliability as **SIB anodes**.

Important Sections' Instructions

3. Research Background

- Should be the research problems cited in the name of the author (i.e. Liu et al [1] stated that lithium batteries contain toxic..blah blah)
- should lead up to your research hypothesis/your project aim

Example:

- Hard carbons are ideal anode materials for sodium-ion batteries due to their “house of cards” microstructure and expanded interlayer spacing, which fit larger Na⁺ ions. [4]
- Sodium storage in hard carbon involves surface adsorption, pore filling, and intercalation between graphene layers, enabling stable cycling. [5]
- **Date seed waste** is an **abundant, eco-friendly** precursor for synthesizing hard carbons with tailored electrochemical properties.
- This work explores **how date seed variety** affects the **structure and performance of hard carbon anodes**.

4. Objectives

- bulleted objectives of the methodology or purpose of your study
- Should start with (to ... i.e. to synthesise x y z novel material for a b c)

Example:

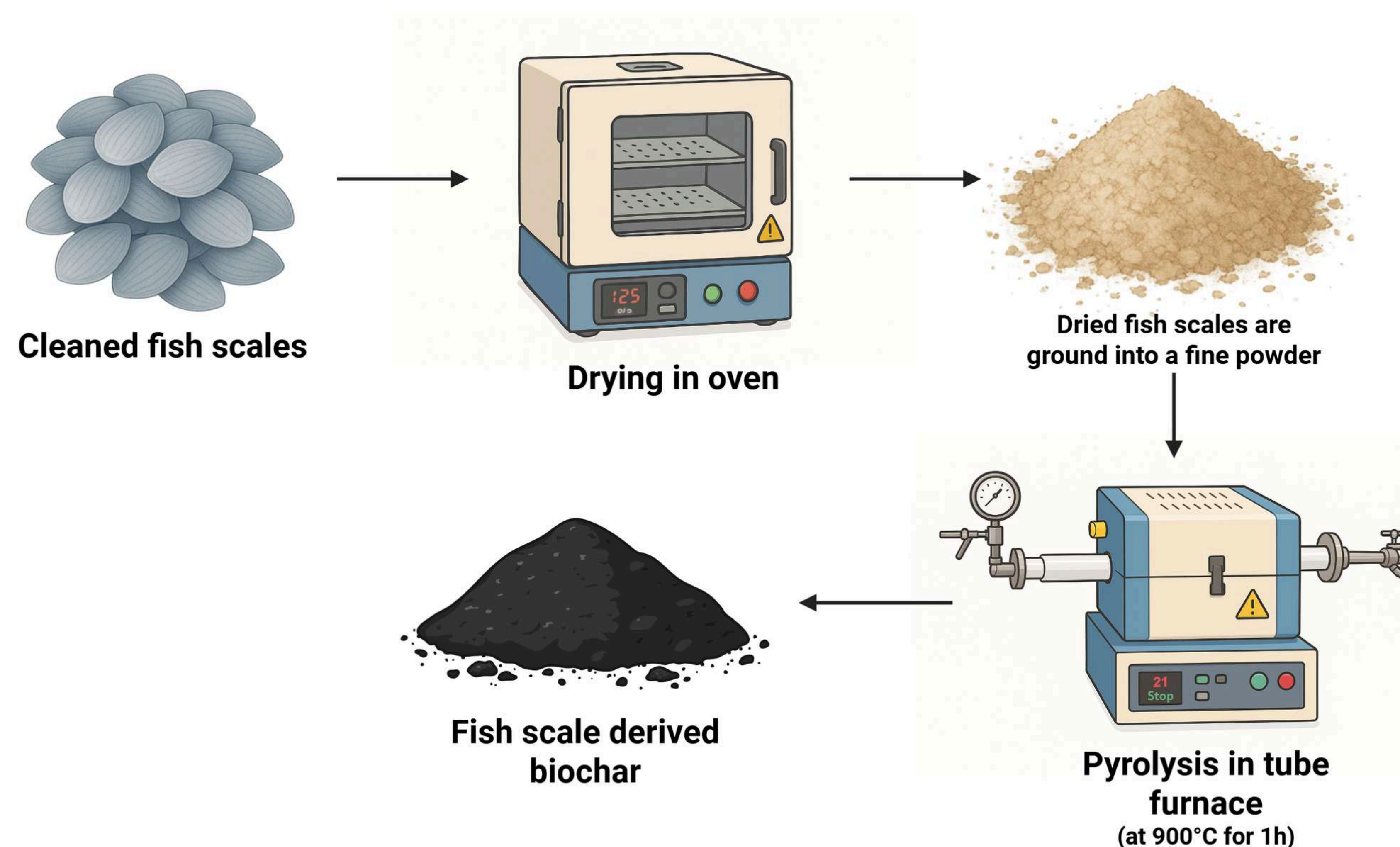
- • To synthesize and characterize biochar derived from fish scales.
- • To evaluate its adsorption efficiency for multiple heavy metals (Pb, Cd, Cr, Ni) in contaminated water.
- • To optimize adsorption conditions (pH, contact time, temperature, biochar dosage).
- • To assess reusability and compare performance with commercial adsorbents.

Important Sections' Instructions

5. Methodology

- Draw the exact steps of your methodology, you can section it depending on your specific requirement.
- Use small comments to enhance the description of the methodology.

Example:



6. Characterization and/or Results

- Add the results after plotting from your mentor in a concise way.
- label each figure underneath every graph (i.e. Fig. 1. FTIR analysis of the sample.)

Example:

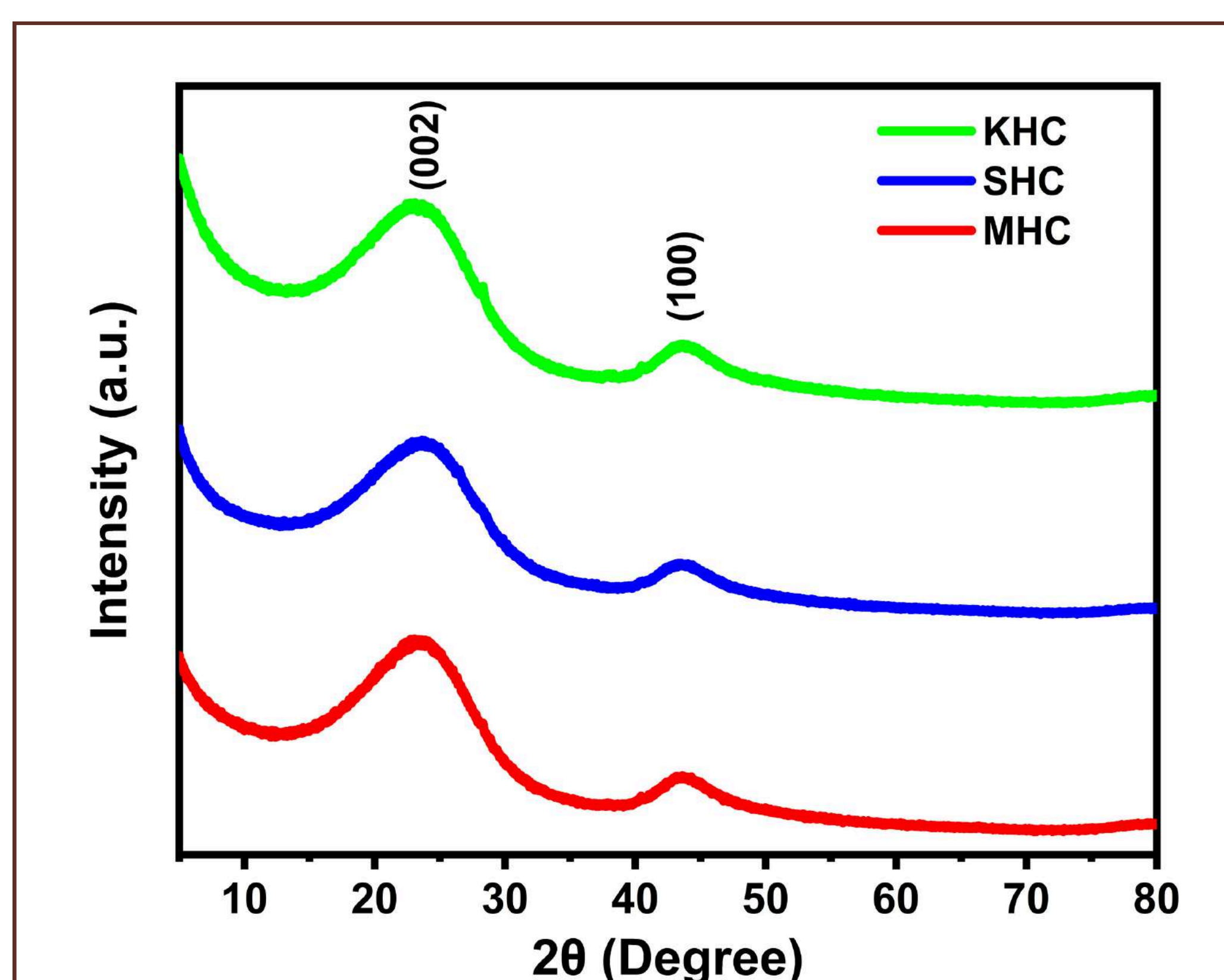


Fig. 1. XRD results of the dates

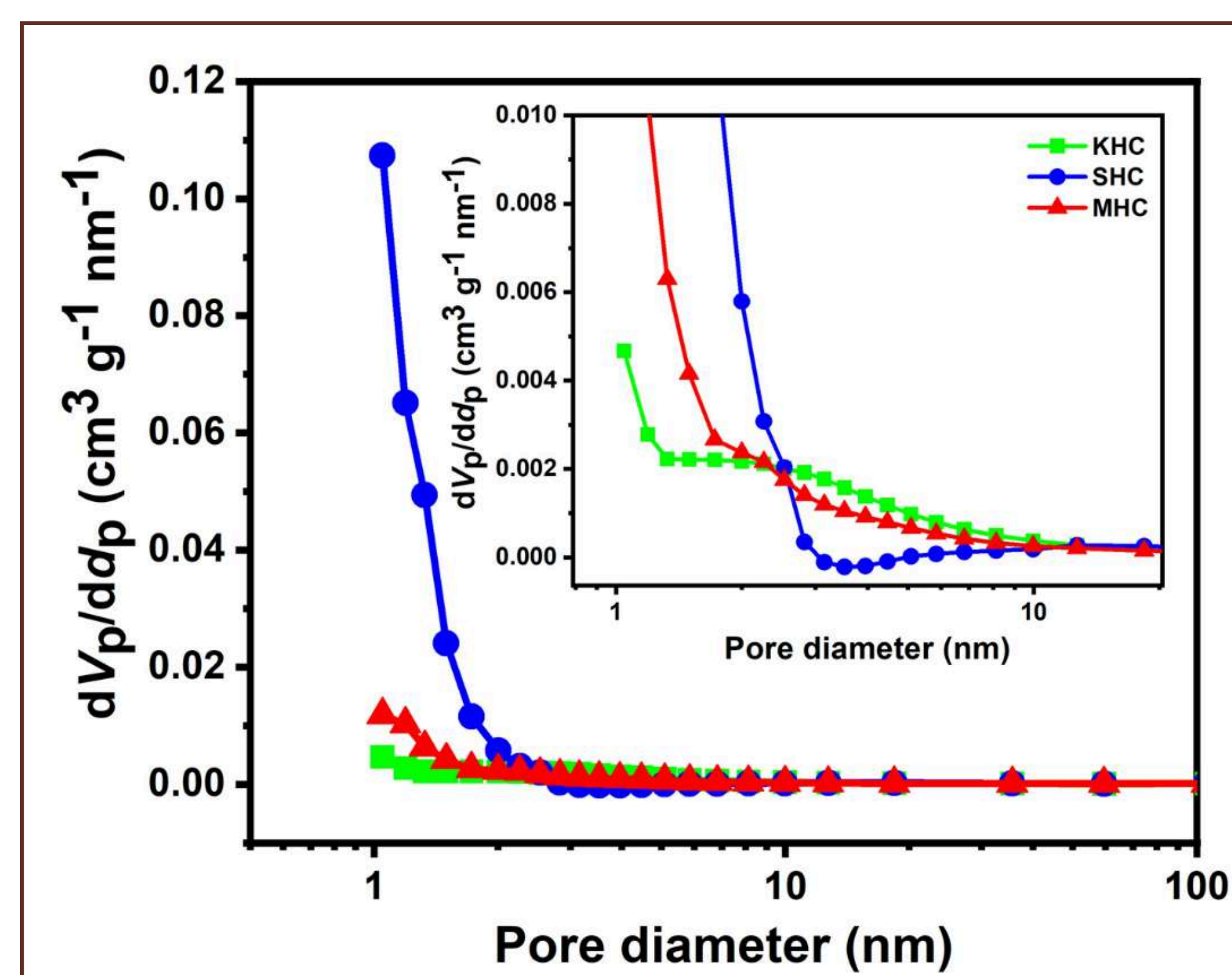


Fig. 2. Pore size distribution.

Important Sections' Instructions

7. Conclusion

- In bullet points, conclude your whole research in main points including the quantitative results of your findings.

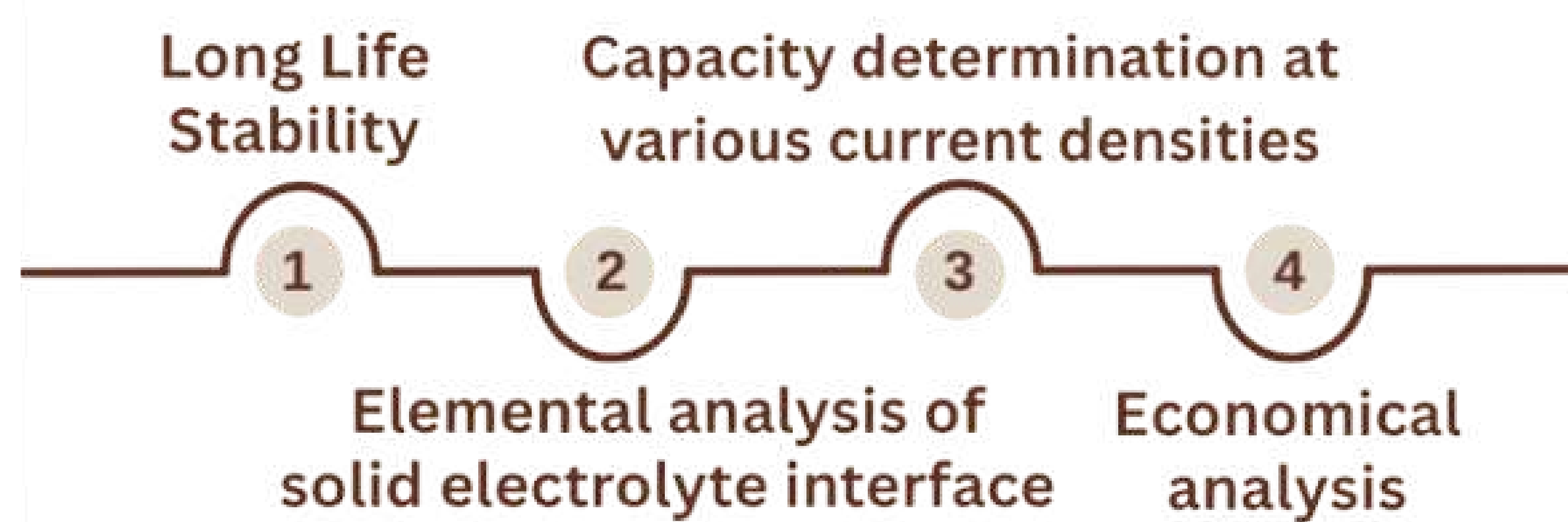
Example:

- All samples demonstrated successful carbonization, minimal impurities (FTIR), and uniform carbon graphitic distribution (EDS).
- All samples show broad peaks at (002), indicating amorphous/disordered hard carbon.
- Khalas, Sukkari, and Mejdool achieved surface areas of $27.6 \text{ m}^2/\text{g}$, $211.1 \text{ m}^2/\text{g}$, and $44.69 \text{ m}^2/\text{g}$ respectively.
- Khalas, Sukkari and Mejdool demonstrated pore sizes of 1.8 nm, 4.8 nm, and 6.3 nm respectively.
- These data illustrate great potential for SIBs applications.

8. Future works

- Explain in small words or in a timeline the exact steps you want to achieve.

Example:



9. Applications & Impact (optional)

- Explain or draw how exactly and where would your project be implemented.
- For example, when synthesising a new formulation of a battery, explain where would it be applied and how (i.e. a battery that's engineered to withstand corrosion may be applied for marines applications in a water system)

Important Sections' Instructions

10. References

- Reference the main papers you read and stated in your poster.



- Lutfi, M., Hanafi, B., Susilo, B., Prasetyo, J., Sandra, and Prajogo, U. (2021). "Characteristics of activated carbon from coconut shell (Cocos nucifera) through chemical activation process." IOP Conference Series: Earth and Environmental Science, 733(1), 012134.
- Keppetipola, N. M., Dissanayake, M., Dissanayake, P., Karunaratne, B., Dourges, M. A., Talaga, D., Servant, L., Olivier, C., Toupance, T., Uchida, S., Tennakone, K., Kumara, G. R. A., & Cojocar, L. (2021). Graphite-type activated carbon from coconut shell: a natural source for eco-friendly non-volatile storage devices. RSC Advances, 11, 2854–2865.
- Sujiono, E. H., Rahayu, F., Haryono, A., Prasetyo, D. B., & Nugroho, W. S. (2022). Fabrication and characterization of coconut shell activated carbon for supercapacitor applications. Results in Chemistry, 4, 100486.
- Saputro, E. A., Wulan, V. D. R., Winata, B. Y., Yogaswara, R. R., & Erliyanti, N. K. (2020). Process of Activated Carbon from Coconut Shells Through Chemical Activation. Natural Science: Journal of Science and Technology, 9(1).
- Akhil Pradiprao Khedulkar, Bidhan Pandit, Van Dien Dang, Ruey-an Doong, Agricultural waste to real worth biochar as a sustainable material for supercapacitor, Science of The Total Environment, Volume 869,2023,161441,ISSN 00489697
- Anand Ramanathan, K.M. Meera Sheriffa Begum, Amaro Olimpio Pereira, Claude Cohen, Chapter 1 - Pyrolysis of waste biomass: toward sustainable development, Editor(s): Anand Ramanathan, K.M. Meera Sheriffa Begum, Amaro Olimpio Pereira, Claude Cohen, A Thermo-Economic Approach to Energy From Waste, Elsevier, 2022, Pages 1-34, ISBN 9780128243572,

- DO state each paper with a number prior
- DO NOT barcode your reference

1

Google Scholar

Lithium-ion batteries anodes

A new way to search

Try Scholar Labs

Stand on the shoulders of giants

Lithium-ion batteries anodes

2

966,000 results (0.17 sec)

Interconnected silicon hollow nanospheres for lithium-ion battery anodes with long cycle life

Y Yao, MT McDowell, I Ryu, H Wu, N Liu, L Hu... - Nano ..., 2011 - ACS Publications

... Silicon is a promising candidate for the **anode** material in **lithium-ion batteries** due to its high theoretical specific capacity. However, volume changes during cycling cause pulverization ...

☆ Save Cite Cited by 1781 Related articles All 17 versions

Cite

3

MLA Yao, Yan, et al. "Interconnected silicon hollow nanospheres for lithium-ion battery anodes with long cycle life." *Nano letters* 11.7 (2011): 2949-2954.

APA Yao, Y., McDowell, M. T., Ryu, I., Wu, H., Liu, N., Hu, L., ... & Cui, Y. (2011). Interconnected silicon hollow nanospheres for lithium-ion battery anodes with long cycle life. *Nano letters*, 11(7), 2949-2954.

Chicago Yao, Yan, Matthew T. McDowell, Ill Ryu, Hui Wu, Nian Liu, Liangbing Hu, William D. Nix, and Yi Cui. "Interconnected silicon hollow nanospheres for lithium-ion battery anodes with long cycle life." *Nano letters* 11, no. 7 (2011): 2949-2954.

Harvard Yao, Y., McDowell, M.T., Ryu, I., Wu, H., Liu, N., Hu, L., Nix, W.D. and Cui, Y., 2011. Interconnected silicon hollow nanospheres for lithium-ion battery anodes with long cycle life. *Nano letters*, 11(7), pp.2949-2954.

Vancouver Yao Y, McDowell MT, Ryu I, Wu H, Liu N, Hu L, Nix WD, Cui Y. Interconnected silicon hollow nanospheres for lithium-ion battery anodes with long cycle life. *Nano letters*. 2011 Jul 13;11(7):2949-54.

BibTeX EndNote RefMan RefWorks

Important Sections' Instructions

11. Header and Official Branding

Before editing your research poster:

- Download and install the provided fonts on your device to prevent any changes to the poster's typography or formatting.
- Use the official poster template assigned to your research track.
- Keep the official logos, their placement, and the project title section exactly as provided in the template.
- **Do not delete, replace, resize, move, crop, or recolor any of the official logos.**
- Add your research title only within the designated project title area. You may adjust the title's font size when necessary, while keeping it clear and readable.
- All other poster sections may be edited, reorganized, or customized according to your research content.

The header shown below must remain consistent in the final submitted poster.



Important Sections' Instructions

12. Labeling

Fig. X. Explain the figure + label and cite the website used

Example

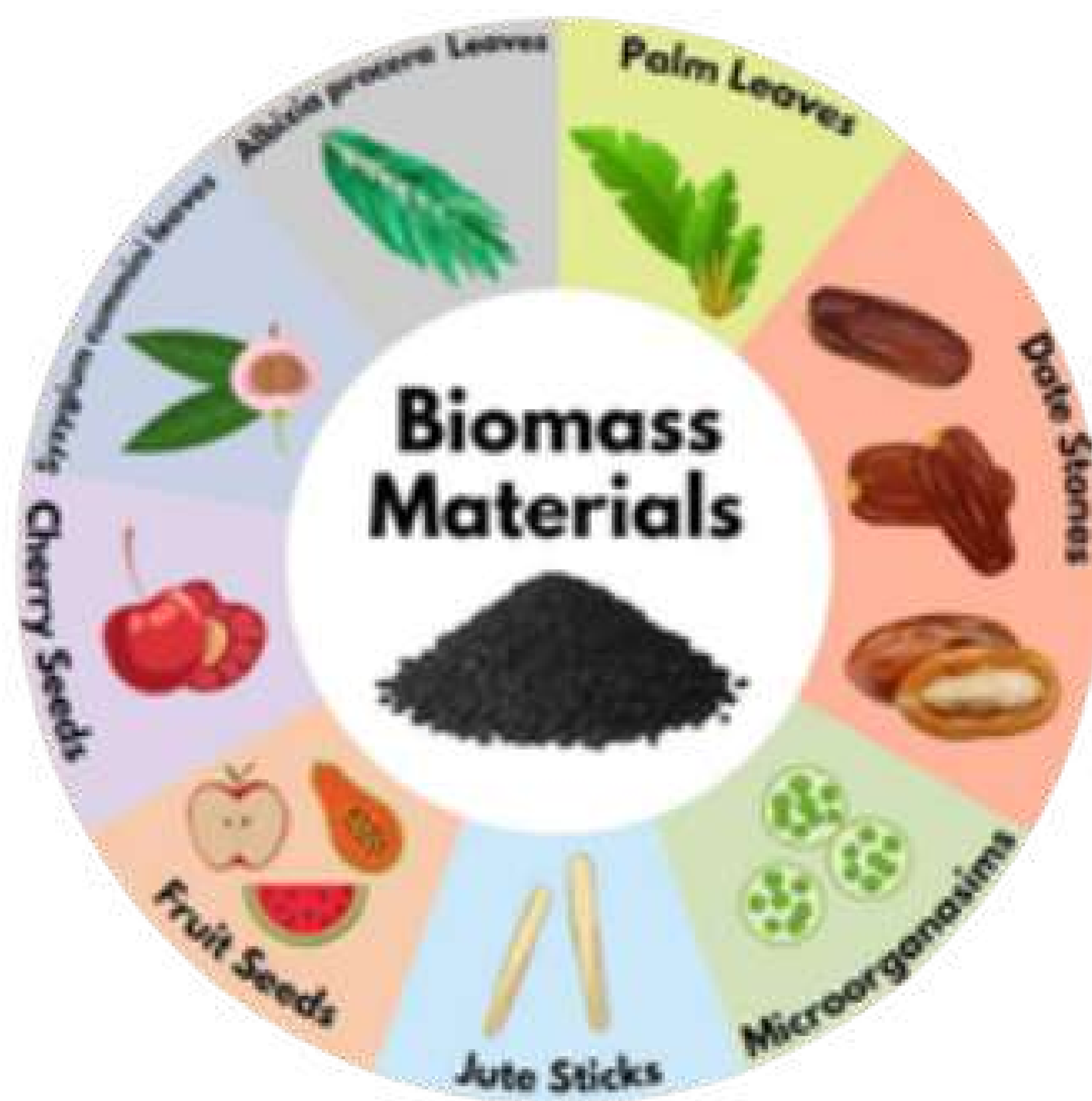


Fig. 1. Illustration of biomass derived materials for sodium ion batteries. Made by student via Canva.com, 2026.

Important Sections' Instructions

13. Professional VS Poor Writing



Sodium-ion batteries have recently received significant attention as alternative energy-storage systems because sodium is widely available and potentially less expensive than lithium. However, their electrochemical performance is strongly influenced by the properties of the anode material, and conventional graphite is generally unsuitable for efficient sodium storage. Hard carbon has therefore emerged as a promising anode material because of its disordered structure, expanded interlayer spacing, defects, and pores. In this study, date seeds were collected, cleaned, dried, ground, carbonised, and characterised using several analytical techniques. The prepared materials were then tested in sodium-ion battery cells to evaluate their capacity, cycling stability, and efficiency. The results demonstrated that date-seed-derived hard carbon could provide promising electrochemical performance, indicating its potential as an affordable and sustainable anode material for future sodium-ion batteries.



- Growing energy demands require affordable and sustainable battery storage technologies.
- Sodium-ion batteries offer abundant resources but need improved anode materials.
- Hard carbon provides disordered structures suitable for reversible sodium storage.
- Date seeds are abundant agricultural waste with high carbon content.
- This study evaluates date-seed hard carbon as a sustainable anode.

Instructions

- Should be written in 4-5 bullets MAX
- Each bullet should have 10-12 words MAX
- Include important statistics and cite the reference used in the "References" section
- This should be applied to ALL the sections

MERP Poster



موهبة الإشراف
البحوثي 2026
جامعة الملك فهد للبترول والمعادن



موهبة
Mawhiba
مؤسسة الملك عبد العزيز
ورعاية الموهبة والإبداع
KING ABDOULAZIZ AND HIS COMPANIONS
FOUNDATION FOR GIFTEDNESS AND CREATIVITY

Unit Name

Project Title

Team Member Names

Introduction

This text is an example of a text that can be replaced in the same space, this text was generated from the Arabic text generator, where you can generate such text or many other texts in addition to increasing the number of characters generated by the application. If you need a larger number of paragraphs, the Arabic text generator allows you to increase the number of paragraphs as you want, the text will not appear divided or contain grammatical errors. The role of the Arabic text generator is to save the designer the trouble of searching for an alternative text that has nothing to do with the topic that the design talks about, so it appears in an inappropriate form. This text can be superimposed on any design without issue and will not look like a copied, unorganized, unformatted, or even unintelligible text. It's still an alternative and temporary text.

Methodology

This text is an example of a text that can be replaced in the same space, this text was generated from the Arabic text generator, where you can generate such text or many other texts in addition to increasing the number of characters generated by the application. If you need a larger number of paragraphs, the Arabic text generator allows you to increase the number of paragraphs as you want, the text will not appear divided or contain grammatical errors. The role of the Arabic text generator is to save the designer the trouble of searching for an alternative text that has nothing to do with the topic that the design talks about, so it appears in an inappropriate form. This text can be superimposed on any design without issue and will not look like a copied, unorganized, unformatted, or even unintelligible text. It's still an alternative and temporary text.

Conclusion

This text is an example of a text that can be replaced in the same space, this text was generated from the Arabic text generator, where you can generate such text or many other texts in addition to increasing the number of characters generated by the application. If you need a larger number of paragraphs, the Arabic text generator allows you to increase the number of paragraphs as you want, the text will not appear divided or contain grammatical errors. The role of the Arabic text generator is to save the designer the trouble of searching for an alternative text that has nothing to do with the topic that the design talks about, so it appears in an inappropriate form. This text can be superimposed on any design without issue and will not look like a copied, unorganized, unformatted, or even unintelligible text. It's still an alternative and temporary text.

This text is an example of a text that can be replaced in the same space, this text was generated from the Arabic text generator, where you can generate such text or many other texts in addition to increasing the number of characters generated by the application. If you need a larger number of paragraphs, the Arabic text generator allows you to increase the number of paragraphs as you want, the text will not appear divided or contain grammatical errors. The role of the Arabic text generator is to save the designer the trouble of searching for an alternative text that has nothing to do with the topic that the design talks about, so it appears in an inappropriate form. This text can be superimposed on any design without issue and will not look like a copied, unorganized, unformatted, or even unintelligible text. It's still an alternative and temporary text.

Summary

This text is an example of a text that can be replaced in the same space, this text was generated from the Arabic text generator, where you can generate such text or many other texts in addition to increasing the number of characters generated by the application. If you need a larger number of paragraphs, the Arabic text generator allows you to increase the number of paragraphs as you want, the text will not appear divided or contain grammatical errors. The role of the Arabic text generator is to save the designer the trouble of searching for an alternative text that has nothing to do with the topic that the design talks about, so it appears in an inappropriate form. This text can be superimposed on any design without issue and will not look like a copied, unorganized, unformatted, or even unintelligible text. It's still an alternative and temporary text.

Results

This text is an example of a text that can be replaced in the same space, this text was generated from the Arabic text generator, where you can generate such text or many other texts in addition to increasing the number of characters generated by the application. If you need a larger number of paragraphs, the Arabic text generator allows you to increase the number of paragraphs as you want, the text will not appear divided or contain grammatical errors. The role of the Arabic text generator is to save the designer the trouble of searching for an alternative text that has nothing to do with the topic that the design talks about, so it appears in an inappropriate form.



References

This text is an example of a text that can be replaced in the same space, this text was generated from the Arabic text generator, where you can generate such text or many other texts in addition to increasing the number of characters generated by the application.

MERP Poster



To get the Poster Sample



MERP Poster



Environmental Sustainability and Basic Needs

Title

Stident Name

Mentor Name

Center Name

Motivation

Background

Objectives

Methodology

Characterization

Results Optimization

Conclusion

Future Work

References

Impact

Example of Creative Poster Template

MERP Poster (Creativity Example)



Environmental Sustainability and Basic Needs

Title

Student Name

Mentor Name

Center Name

Motivation

Background

Objectives

Methodology

Characterization

Results Optimization

Conclusion

Future Work

References

Impact

Poster Link

